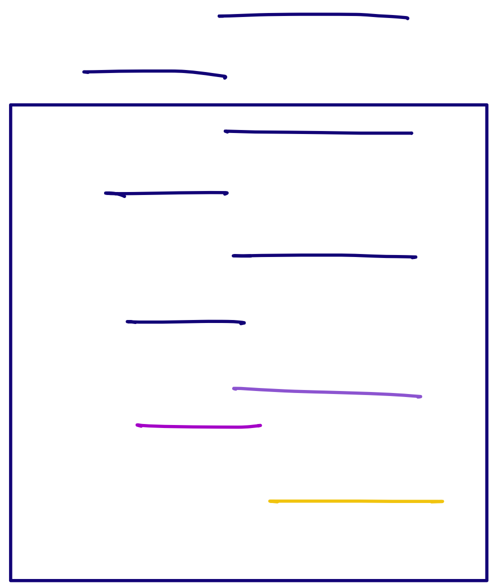
Prompt Engineering

↓

Art of writing prompt, Choosing the right words to guide the model to get high quality output.



⇒ Elements of prompt

1) Instruction
2) Context
3) Constraints
4) Format
5) Variables.

→ Prompt : Generate a {x} days itinery from {source} to {dest}, ______

{variable_name}

Parameters of a Prompt

↳ Used to control LLM's performance.

- Temperature
- Sampling
- Repitition Penalty
- Max Tokens.

Temperature. → 0 to 2

↳ Controls randomness & creativity of the response.

low temperature ⇒ low creativity OR More confident answers.

↓

Reasoning | Research | Maths | Coding

high temperature ⇒ high creativity

↳ Creative work.

```
llm = OpenAI()

llm.responses.create(
    model = "gpt-5.5",
    prompt = ___,
    temperature = 1.1
```

https://github.com/kaseradeepak/BookSummarizer

Sampling

→ LLM's work by predicting the probabilities of next possible tokens & then sampling the highest from them to generate the final response.

- → Top P
- → Top K

Top-P.

↳ Works by selecting the tokens from the cumulative probability distribution until a certain threshold is met.

Query : ____

LLM

P = 0.5.

T_1 :	0.15	0.15
T_2 :	0.11	0.26
T_3 :	0.09	0.35
T_4 :	0.07	0.42
⋮	0.06	0.48
⋮	0.04	0.52
	≡	≡

1 ← Total probability.

Top-k.

↳ Selects the token from top-k as per the probability distribution.

Repitition Penalty

↳ Controls the probability of generating tokens that have already been generated.

→ default = 1.0

Higher value of repitition penalty ⇒ Reduce Repitition.

Max Tokens.

↳ Controls the no. of tokens in the response.

Introduction to Memory in AI Applications.

→ Memory vs Context Window.

↓

Amount of text | tokens an LLM can process at a time.

Stateless Agents.

↓

Stateless Agent doesn't retain any information from previous conversations.

Pros.

- Easy to build
- Simple architecture
- Low memory requirement.

Cons:

- Zero personalisation.

Stateful Agents.

→ Retains information about the user over time & use it for future iterations.

Pros

Cons

- Complex architecture
- Cost